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Homework

A homework for class **MET CS 579** at Boston University, instructed by **Jae Lee**. Student Jin Yu Zhang submitted on **March 28, 2023**.

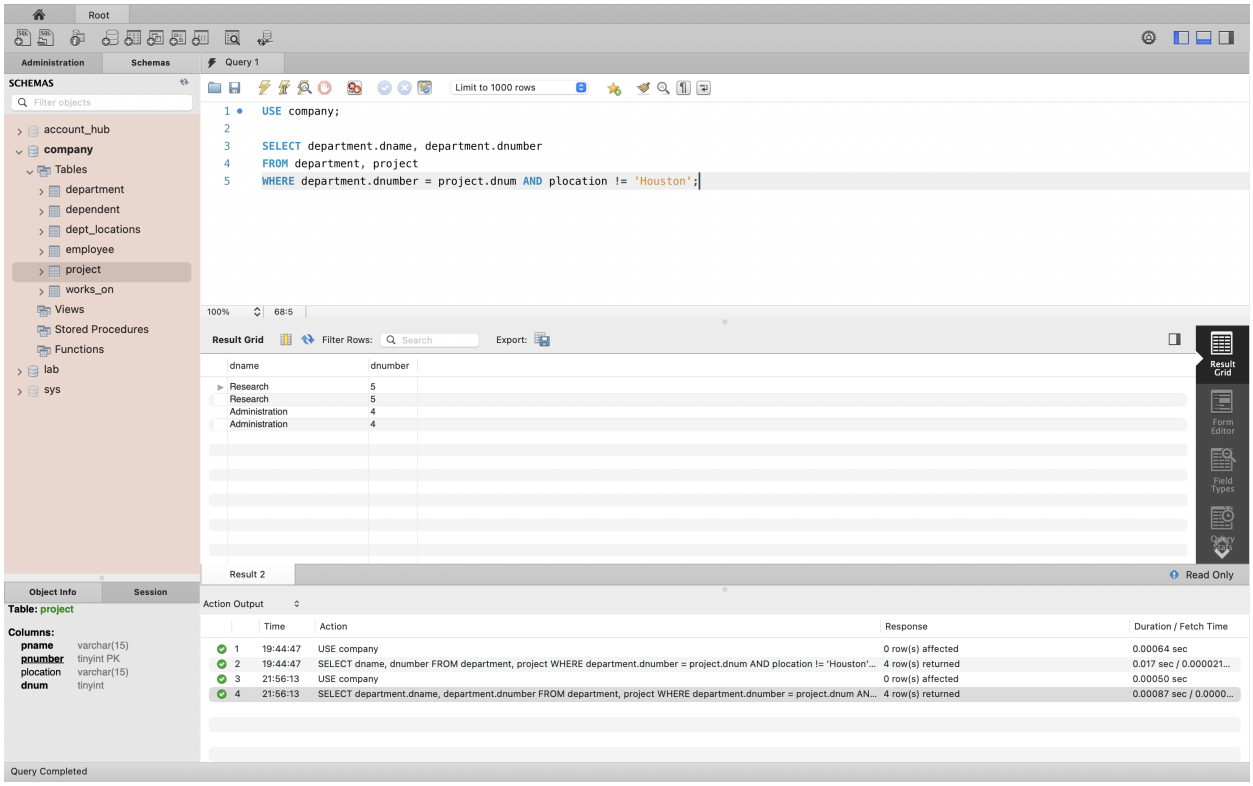
Problem 1

1

Show the department number and the department name of every department that controls a project that is not located in Houston:

```
USE company;

SELECT department.dname, department.dnumber
FROM department, project
WHERE department.dnumber = project.dnum AND plocation != 'Houston';
```

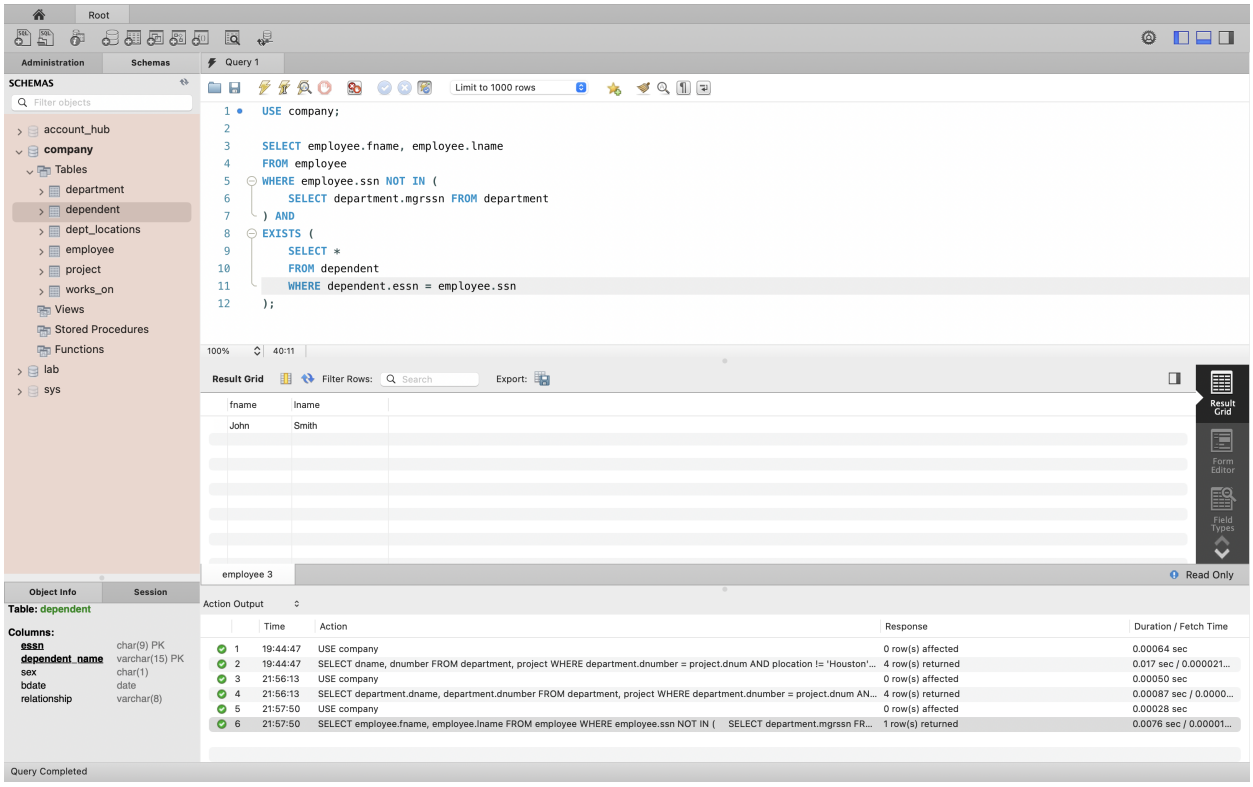


2

Show the first name and the last name of every employee who is a not manager and has a dependent:

```
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn NOT IN (
    SELECT department.mgrssn FROM department
) AND
EXISTS (
    SELECT *
    FROM dependent
    WHERE dependent.essn = employee.ssn
);
```



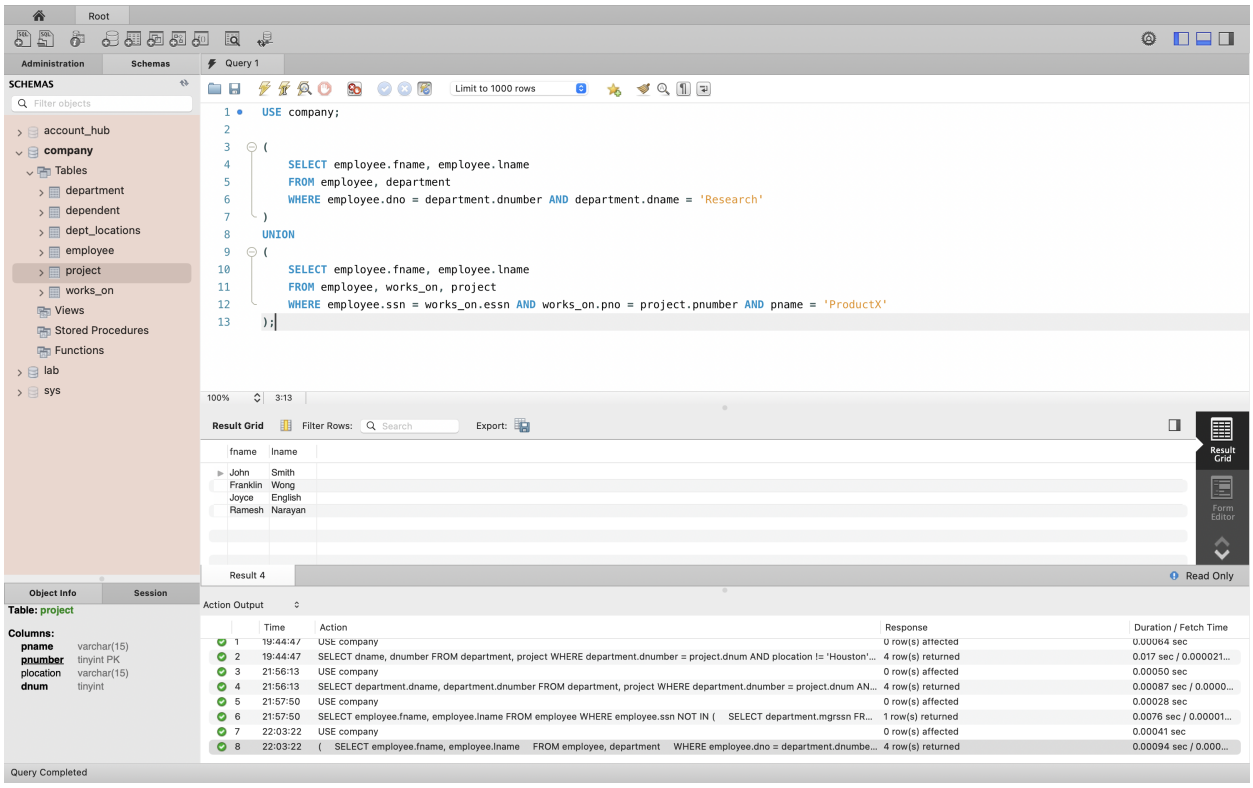
3

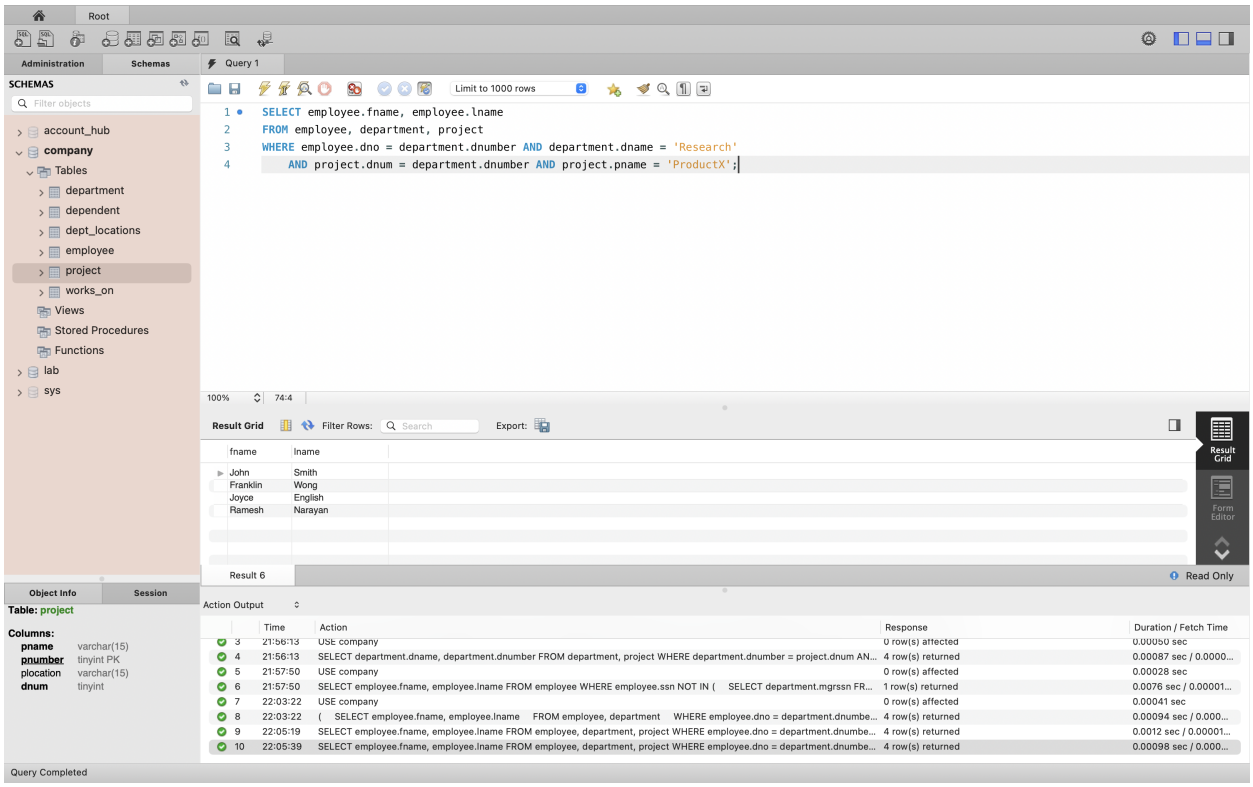
Show the first name and the last name of every employee who works for the Research department and works on the ProductX project:

```
USE company;

SELECT employee.fname, employee.lname
```

```sql
FROM employee, department, project
WHERE employee.dno = department.dnumber AND department.dname = 'Research
    AND project.dnum = department.dnumber AND project.pname = 'ProductX'
-- or (I feel using UNION is more readable, but the first one seems to b
(
    SELECT employee.fname, employee.lname
    FROM employee, department
    WHERE employee.dno = department.dnumber AND department.dname = 'Rese
)
UNION
(
    SELECT employee.fname, employee.lname
    FROM employee, works_on, project
    WHERE employee.ssn = works_on.essn AND works_on.pno = project.pnumbe
);
```





4

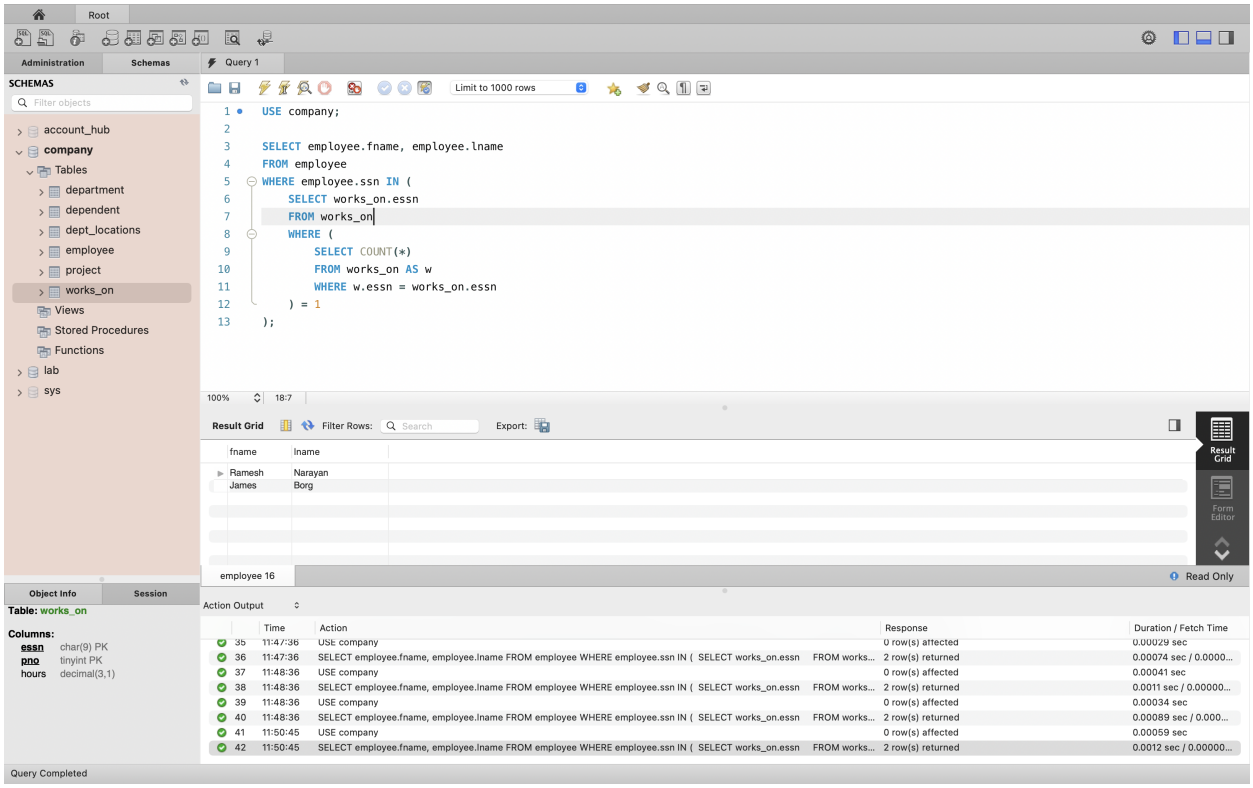
Show the first name and last name of every employee who works on only one project:

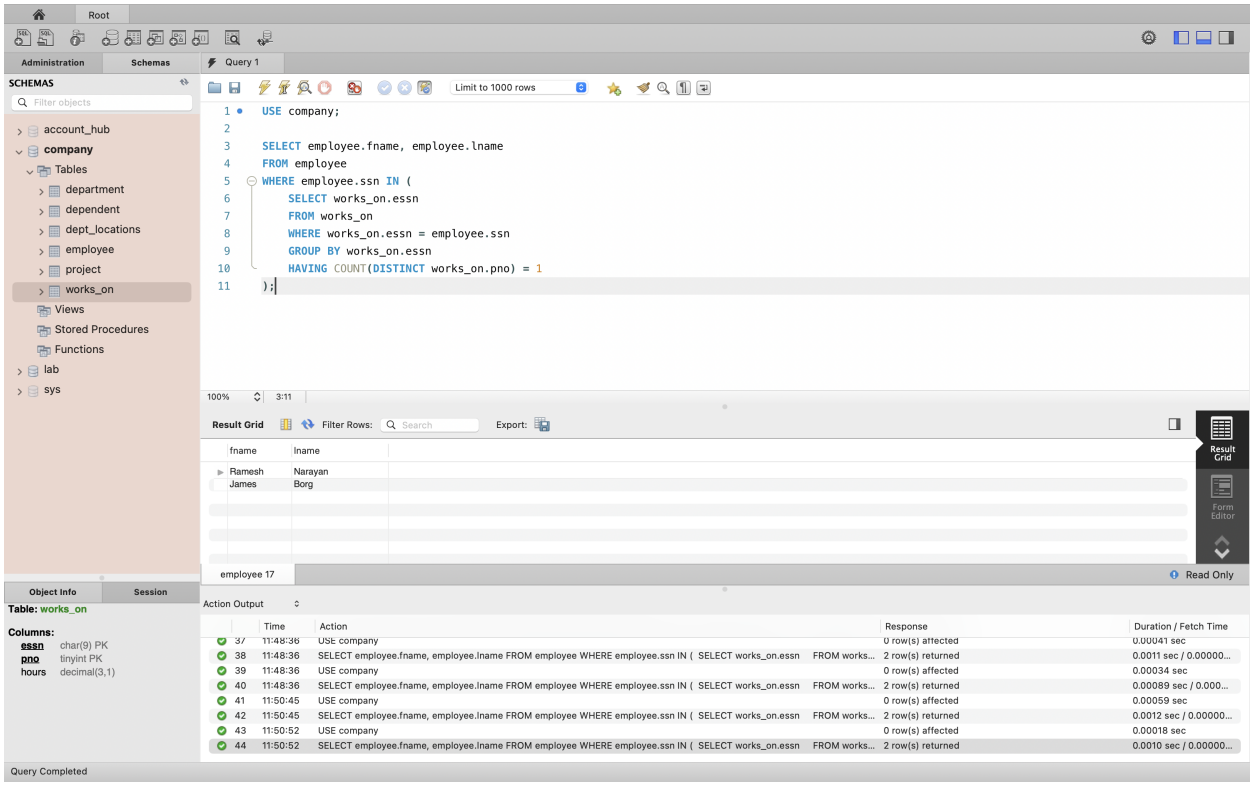
```sql
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn IN (
    SELECT works_on.essn
    FROM works_on
    WHERE (
        SELECT COUNT(*)
        FROM works_on AS w
        WHERE w.essn = works_on.essn
    ) = 1
);
-- or
USE company;

SELECT employee.fname, employee.lname
FROM employee
WHERE employee.ssn IN (
    SELECT works_on.essn
    FROM works_on
    WHERE works_on.essn = employee.ssn
```

```
    GROUP BY works_on.essn
    HAVING COUNT(DISTINCT works_on.pno) = 1
);
```





5

Show the first name and the last name of every employee. If an employee is the manager of a department, then show the name of the department the employee manages too. The result must

be in a single relation:

```
USE company;

SELECt employee.fname, employee.lname, department.dname
FROM employee LEFT JOIN department ON employee.ssn = department.mgrssn;
```

